

Computer Graphics and Image Processing Laboratory

Project Proposal

Roll No.	2107064
Project Title	Moonlit Village Gathering – A Rural Night Scene in 3D

1. Project Description

The project recreates a nostalgic rural Bangladeshi summer night during a power outage, when a family gathers in an open courtyard under full moonlight. On a woven charpai, parents chat and fan themselves with a handmade wooden fan; nearby, children run about playing hide-and-seek, while older siblings sit reading a Bengali poem aloud. Fireflies blink around the yard, and everything is lit only by the moon.

Every object in the scene is generated procedurally in OpenGL 3.3 core-profile code from mathematical primitives (spheres, cylinders, cones, planes) — no external models are used. A single directional “moon” light drives per-pixel Phong shading (ambient + diffuse + specular) across all surfaces, while the scene demonstrates a wide range of hierarchical and time-varying transformations: the spinning fan assembly, walking/hiding children, a swaying tree canopy, and blinking fireflies. The animation is organized into clear stages (settling in → fan swaying → hide-and-seek → poem reading → firefly glow) rather than one continuous unstructured loop, matching the lab’s requirement for meaningful scene structure.

2. Objects Used

- **Terrain:** ground built from a subdivided procedural grid (parametric plane equation), tinted dark grey-green for a moonlit yard.
- **Charpai (cot):** legs generated as cylinders (parametric radius–height sweep); woven top surface as a criss-cross grid of thin cuboids for the rope-weave look.
- **Handmade fan:** hub as a cylinder; 3–4 blades as flat rotated rectangles parented to the hub — classic hierarchical parent–child rotation.
- **Family figures:** torso/limbs/head as stacked spheres and cylinders (procedural humanoid rig) for parents, older siblings, and running children.
- **Books:** thin flattened cuboids held by the reading siblings, angled toward the moonlight.
- **Trees / plants:** trunk as a cylinder; canopy as clustered spheres/cones, positioned procedurally around the yard boundary.
- **Fireflies:** small emissive spheres with sine-based blink/opacity animation, scattered using a procedural random-placement algorithm.
- **Moon:** a large sphere placed far in the sky, doubling as the visual source for the directional light.
- **Hut / house silhouette (background):** simple cylinder/cone assemblies scattered as background silhouettes to frame the courtyard.

3. Extra Features (Beyond Baseline Requirement)

- **Multi-part rigged characters:** procedural humanoid figures for family members/children, giving extra hierarchical joints (shoulder → arm, hip → leg) beyond the minimum required transformation set.
- **Glow / emissive shading:** fireflies use a pulsing emissive term layered on top of Phong shading to fake a glow without a full bloom pass.
- **Instanced-style rendering:** fireflies and leaf clusters are drawn from one shared mesh (VBO/EBO) reused across many instances with only the model matrix changed per draw — reduces memory and draw-call overhead.
- **Stage-based scene director:** a lightweight scene-state machine cross-fades between the five narrative stages using a full-screen alpha overlay, clearly separating them as requested.
- **Wind-sway animation:** tree canopies and fan blades get a small sinusoidal rotation/offset over time, adding secondary motion without new geometry.
- **Procedural scene population:** tree and plant placement uses a seeded pseudo-random procedural layout so the yard never looks perfectly symmetrical.
- **Camera fly-through:** a slow orbital camera pass over the courtyard is included so all lighting/transformation cases (fan, fireflies, figures) are visible from multiple angles.